



Indian Institute of Science Education and Research Thiruvananthapuram
IDC 222 - Scientific Computing and Data Visualization
(Question Paper Template)

Time: **1 Hr 30 Min.**

Max. Marks: **40**

(Answer all the following questions)

1. Fill in the blanks OR MCQ

2. Pandas { Series or DataFrame }

Given the dataset `car_crashes.csv`, Write a Python program to:

- Load the dataset using **Pandas**.
- Find the minimum value of total crashes (**total**).
- Count how many states have speeding-related crashes (**speeding**) greater than 5.

3. Visualization

Write a Python program to:

- Create a **scatter plot** showing:
 - a. x-axis → **speeding**
 - b. y-axis → **total**
- Add a suitable title to the plot.

4. SciPy {Integration}

A pollutant concentration in a river is given by $C(t) = e^{-0.3t} \sin(t)$

Write a Python program using SciPy to:

- Compute the total pollutant from **$t=0$ to $t=10$** using numerical integration.
- Estimate the error of the integration.
- Print both the result and the error.